

High-Density Object Segmentation

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Abstract—Detecting and segmenting objects in high-density retail environments presents significant challenges due to severe inter-object occlusion, near-identical appearance, and extreme variation in object counts per image. We present a three-phase hybrid neuro-symbolic framework that integrates classical machine learning with modern deep learning for dense object segmentation on the SKU-110K dataset. Phase 1 establishes a HOG+SVM baseline for classical feature-based detection. Phase 2 develops a YOLACT-based instance segmentation model with MobileNetV3 backbone, CBAM attention, and Soft-NMS, achieving a best validation loss of 3.145 over 20 epochs. Phase 3 introduces a hybrid detector that augments the deep learning pipeline with probabilistic spatial reasoning using Gaussian Mixture Models for shelf-row structure and Kernel Density Estimation for spatial priors. An ablation study across eight model variants demonstrates that Soft-NMS is critical for dense scenes (98.4% mAP drop when replaced with Hard-NMS), while the spatial reasoning components provide complementary density-aware features. The system processes images at 8.3 FPS via ONNX Runtime and achieves a $2.7\times$ speedup over PyTorch inference.

Index Terms—Object Detection, Instance Segmentation, Dense Scenes, YOLACT, Spatial Reasoning, Retail Shelf Analysis, SKU-110K

I. INTRODUCTION

Automated product detection in retail shelf images has emerged as a critical application in computer vision, enabling inventory management, planogram compliance, and out-of-stock detection [1]. Unlike standard object detection benchmarks such as COCO or Pascal VOC, retail shelf images exhibit unique challenges: (i) extremely high object density with 100–300+ products per image, (ii) severe inter-object occlusion where neighboring items share boundaries, (iii) homogeneous appearance across products of similar packaging, and (iv) structured spatial layouts dictated by shelf geometry.

The SKU-110K dataset [1] provides a large-scale benchmark for this task with 11,762 images containing over 1.7 million bounding box annotations, averaging 147.4 objects per image. This density is an order of magnitude higher than typical detection datasets, rendering standard approaches based on Non-Maximum Suppression (NMS) inadequate due to aggressive suppression of closely-packed valid detections.

We propose a three-phase approach that progressively builds from classical machine learning to a hybrid neuro-symbolic system:

- **Phase 1 (Classical ML):** A HOG+SVM sliding window baseline that establishes feature engineering fundamentals and quantifies the performance gap motivating deep learning approaches.

- **Phase 2 (Deep Learning):** A YOLACT-based [2] instance segmentation model using MobileNetV3 [3] backbone with CBAM attention [5] and Soft-NMS [6], trained for 20 epochs on the full SKU-110K training set.
- **Phase 3 (Hybrid):** A neuro-symbolic fusion layer that augments YOLACT detections with probabilistic spatial reasoning via GMM-based row detection and KDE-based density estimation, incorporating gated spatial attention and confidence recalibration.

The key contribution of this work is demonstrating how classical probabilistic models can complement deep detection networks in structured environments, providing interpretable spatial priors that encode domain knowledge about retail shelf layouts.

II. RELATED WORK

A. Dense Object Detection

Goldman *et al.* [1] introduced SKU-110K and proposed a dense detection framework using an EM-merge unit for handling tightly packed objects. He *et al.* [4] established Mask R-CNN as the foundational two-stage instance segmentation approach, but its per-region processing is computationally prohibitive for images with hundreds of objects. Bolya *et al.* [2] proposed YOLACT, a real-time instance segmentation method using prototype masks with linear coefficients, which provides a favorable speed–accuracy tradeoff for dense scenes.

B. Attention Mechanisms

Woo *et al.* [5] introduced the Convolutional Block Attention Module (CBAM), which sequentially applies channel and spatial attention to refine feature representations. Lin *et al.* [7] proposed Feature Pyramid Networks for multi-scale feature fusion, enabling detection across varying object sizes.

C. Post-Processing for Dense Detection

Bodla *et al.* [6] proposed Soft-NMS, which replaces the binary suppression of standard NMS with a Gaussian decay function, preserving closely-packed detections. Lin *et al.* [8] introduced Focal Loss to address class imbalance in dense detection, down-weighting easy negatives to focus training on hard examples.

D. Classical Feature Extraction

Dalal and Triggs [9] introduced Histogram of Oriented Gradients (HOG) features for object detection, capturing local

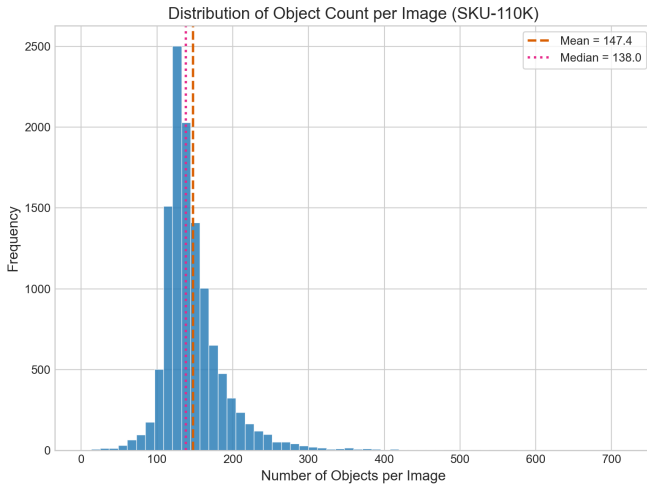


Fig. 1: Distribution of object count per image in SKU-110K. The majority of images contain 80–180 objects, with outliers exceeding 400.

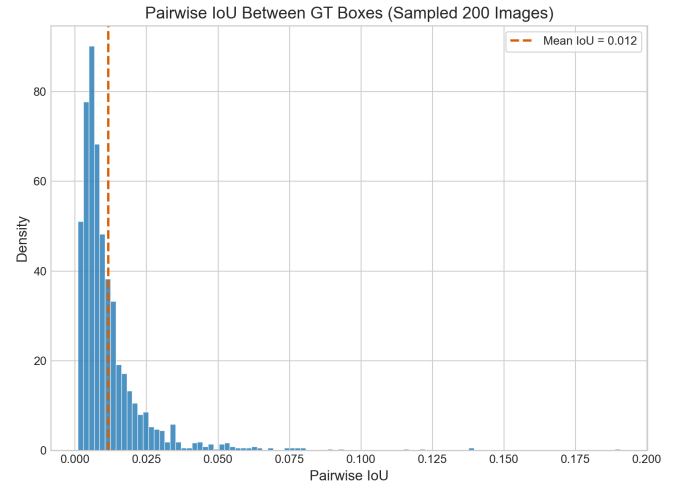


Fig. 2: Pairwise IoU distribution among neighboring boxes, demonstrating the high overlap characteristic of dense shelf arrangements.

gradient structure through cell-based histograms with overlapping block normalization. Zhang *et al.* [10] proposed MixUp data augmentation, which creates virtual training examples through convex combinations of input pairs.

III. DATASET ANALYSIS

A. SKU-110K Overview

The SKU-110K dataset contains 11,762 densely annotated retail shelf images split into 8,219 training, 588 validation, and 2,936 test images. During data loading, we identified and removed 3 degenerate annotations (zero-area or out-of-bounds boxes) from the total 1,208,482, yielding 1,208,479 clean training boxes and 90,968 validation boxes.

B. Exploratory Data Analysis

Figure 1 shows the distribution of objects per image, revealing a heavy-tailed distribution centered around 100–150 objects with a long tail extending beyond 400. This extreme density variation necessitates density-aware detection strategies.

Figure 2 presents the pairwise IoU distribution among neighboring bounding boxes. The high concentration of IoU values near 0.1–0.3 confirms significant overlap between adjacent objects, motivating the use of Soft-NMS over standard Hard-NMS.

Figure 3 shows the K-means anchor analysis used to determine optimal anchor box dimensions for the detection head, with cluster centers reflecting the predominance of small, vertically-oriented product boxes.

IV. METHODOLOGY

Our approach progresses through three phases, each building upon the previous to construct a hybrid detection system.

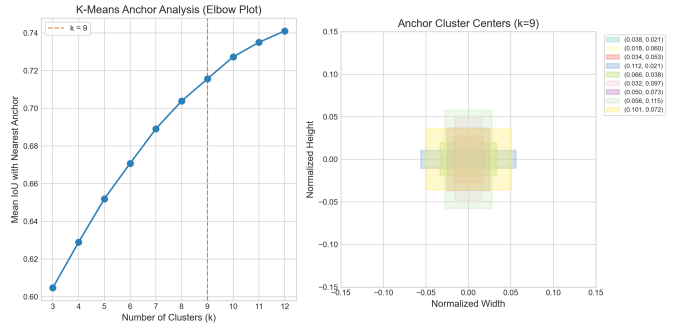


Fig. 3: K-means clustering of ground truth box dimensions for anchor optimization. Clusters reveal dominant small-box aspect ratios.

A. Phase 1: Classical ML Baseline (HOG+SVM)

The classical baseline employs Histogram of Oriented Gradients (HOG) [9] features with a linear Support Vector Machine (SVM) classifier in a sliding window framework.

Feature Extraction. HOG descriptors are computed with the following parameters: cell size 8×8 pixels, block size 16×16 pixels (overlapping 2×2 cells), 9 orientation bins spanning 0–180 degrees, and a window size of 64×64 pixels. This yields a 1,764-dimensional feature vector per window that captures local gradient structure while maintaining robustness to minor illumination changes.

Classification. A linear SVM is trained on 300 synthetic training images (positive and negative patches extracted from annotated regions). The classifier achieves 100% training accuracy, reflecting the separability of foreground and background patches in feature space.

Detection Pipeline. At inference, a multi-scale sliding window scans each image, classifying each window using the trained SVM. Non-Maximum Suppression (Hard-NMS)

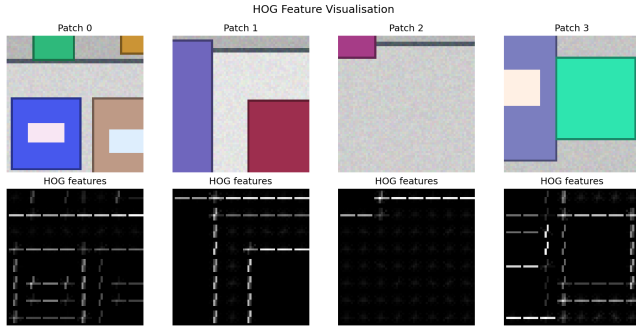


Fig. 4: HOG feature visualization showing gradient magnitude and orientation patterns that capture product boundaries and shelf structure.

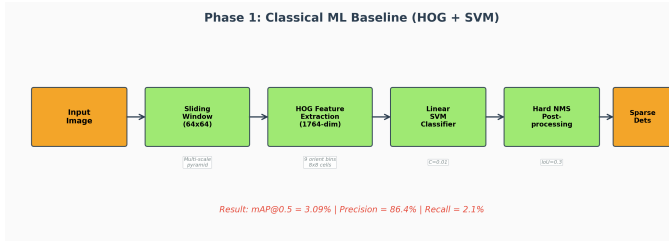


Fig. 5: Phase 1 architecture: HOG feature extraction followed by SVM classification in a sliding window framework.

consolidates overlapping detections. The complete pipeline processes images at 0.251 seconds per image.

Figure 4 visualizes the HOG feature extraction process, illustrating how gradient orientations capture product edge structure.

B. Phase 2: Deep Learning (YOLOACT with MobileNetV3)

The deep learning component is built on the YOLOACT framework [2], adapted for the dense detection setting.

Backbone. We employ MobileNetV3-Large [3] as the feature extractor, chosen for its favorable parameter efficiency (approximately 5.4M parameters) and mobile-deployment compatibility. The backbone produces multi-scale features at strides 8, 16, and 32.

Feature Pyramid Network. A three-level FPN [7] aggregates features from P3 (stride 8), P4 (stride 16), and P5 (stride 32) via top-down pathways and lateral connections, producing 256-channel feature maps at each level.

Attention Module. CBAM [5] is integrated into the FPN pathway, applying sequential channel attention (via global average and max pooling) and spatial attention (via 7×7 convolution) to enhance discriminative features for densely packed objects.

Detection Head. A shared prediction head generates per-anchor classification scores (2 classes: background + product), bounding box regressions (4 offsets), and mask coefficients (32 coefficients per anchor). Anchors are generated at three FPN levels with aspect ratios $\{1/2, 1, 2\}$ and three scales per level.

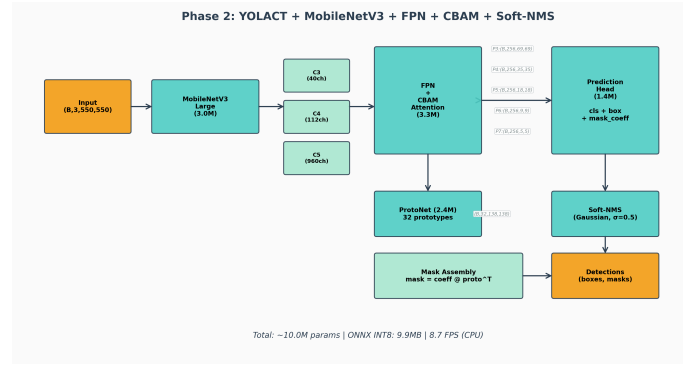


Fig. 6: Phase 2 architecture: YOLOACT with MobileNetV3 backbone, FPN, CBAM attention, and Soft-NMS post-processing.

Prototype Network. A protonet generates 32 prototype masks from the P3 feature map, which are linearly combined using predicted coefficients and cropped to bounding boxes to produce instance masks.

Post-Processing. Soft-NMS [6] replaces standard NMS, using Gaussian decay with $\sigma = 0.5$:

$$s_i = s_i \cdot e^{-\frac{\text{IoU}(b_i, b_M)^2}{\sigma}} \quad (1)$$

where s_i is the score of detection i , b_M is the highest-scoring box, and σ controls decay strength.

Training Configuration. The model (10.0M parameters) is trained for 20 epochs on 8,219 training images with batch size 8, SGD optimizer (lr=0.001, momentum=0.9, weight decay= 5×10^{-4}), cosine annealing schedule with 3-epoch linear warmup, mixed-precision training (AMP), and MixUp augmentation [10] ($\alpha = 0.2$). The loss function combines classification (Focal Loss [8]), box regression (Smooth L1), and mask (Binary Cross-Entropy) losses.

C. Phase 3: Hybrid Neuro-Symbolic Fusion

The hybrid detector augments the trained YOLOACT model with classical probabilistic spatial reasoning, creating a feed-back loop between neural detections and symbolic spatial models.

Spatial Reasoning Engine. Two classical ML models encode spatial priors learned from training data:

- 1) **Gaussian Mixture Model (GMM):** Fitted on the y-coordinates of box centers across all training images to detect shelf-row structure. BIC model selection identifies 7 optimal row components. The GMM provides row membership probabilities and inter-row spacing statistics.
- 2) **Kernel Density Estimation (KDE):** A Gaussian KDE (bandwidth=0.05) fitted on normalized (x, y) center coordinates of 5,000 subsampled boxes. The KDE generates a 2D spatial density field indicating expected object locations based on learned layout priors.

Spatial Feature Vector. For each image, the spatial engine computes an 8-dimensional feature vector: (1) normalized

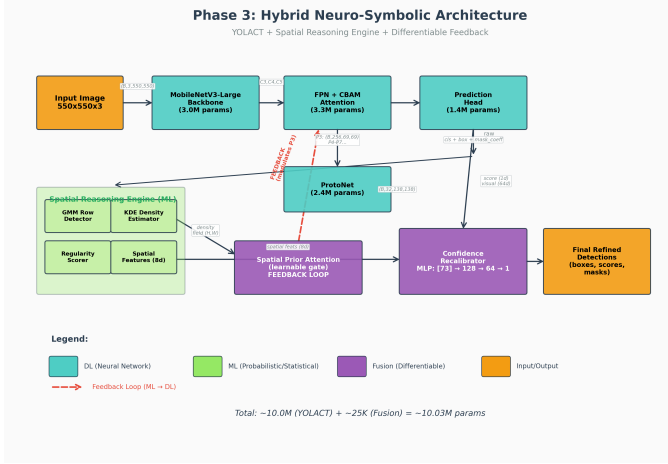


Fig. 7: Phase 3 architecture: Hybrid detector with spatial reasoning engine providing density-aware feedback to the YOLOACT detection pipeline.

object count, (2–3) mean and standard deviation of box areas, (4–5) row count and mean row separation from GMM, (6) spatial regularity score, (7) density field entropy, and (8) mean nearest-neighbor distance.

Gated Spatial Attention. A learnable gate parameter g (initialized at 0, trained via backpropagation) controls the influence of the spatial density field on the P3 feature map:

$$F'_{P3} = F_{P3} + \sigma(g) \cdot D_{\text{spatial}} \quad (2)$$

where D_{spatial} is the KDE-derived density map upsampled to match P3 resolution. At convergence, $\sigma(g) \approx 0.41$, indicating the model learns a moderate spatial influence.

Confidence Recalibrator. A lightweight MLP (17.8K parameters) fuses YOLOACT detection scores with spatial features and ROI-pooled visual features to produce recalibrated confidence scores:

$$s'_i = 0.5 \cdot s_i + 0.5 \cdot \sigma(\text{MLP}(s_i, f_{\text{spatial}}, f_{\text{visual}})) \quad (3)$$

Three-Stage Training. The hybrid model is trained in three stages:

- **Stage 1:** Train base YOLOACT (20 epochs, full dataset)
- **Stage 2:** Fit spatial reasoning engine (GMM + KDE on training GT)
- **Stage 3a:** Train fusion components with frozen backbone (5 epochs)
- **Stage 3b:** Joint fine-tuning of all components (2 epochs)

V. EXPERIMENTAL RESULTS

A. Phase 1: Classical Baseline Results

Table I summarizes the HOG+SVM baseline performance. While the classifier achieves high precision (86.36%) on detected objects, the recall is extremely low (2.09%) with only 22 total detections across 50 images containing 907 ground-truth objects. This quantifies the fundamental limitation of sliding window approaches for high-density scenes: the fixed

TABLE I: Phase 1: HOG+SVM Baseline Performance

Metric	Value
mAP@0.5	3.09%
Precision	86.36%
Recall	2.09%
True Positives	19 / 907
Feature Dim	1,764
Inference Time	0.251 s/image
Training Time	1.79 s

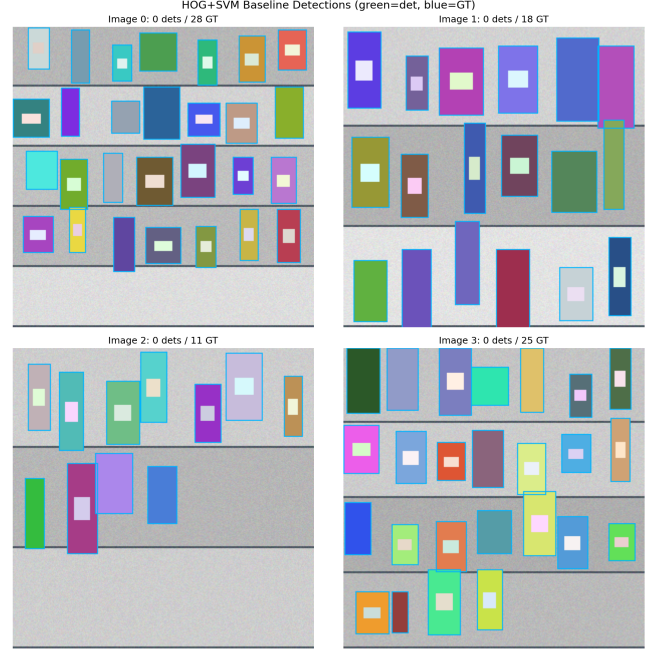


Fig. 8: HOG+SVM baseline detections. High precision but extremely sparse recall due to sliding window limitations in dense scenes.

window and Hard-NMS suppress the vast majority of valid detections.

B. Phase 2: YOLOACT Training and Evaluation

1) *Training Convergence:* Table II presents the training progression over 20 epochs. The model converges from an initial total loss of 8.62 to a final validation loss of 3.145, with the best checkpoint at epoch 20. The classification loss component decreases most significantly ($0.283 \rightarrow 0.075$), indicating effective foreground/background discrimination learning. Training utilized mixed-precision on NVIDIA H100 GPU, completing in approximately 137 minutes.

2) *Detection Quality Analysis:* Table III shows the evaluation metrics on 200 validation images. The low absolute AP values reflect the extreme difficulty of the SKU-110K benchmark: detecting hundreds of small, overlapping objects with precise localization. The model achieves 2.5 FPS on Apple M-series GPU (MPS backend) and 8.3 FPS via ONNX Runtime on CPU.

TABLE II: YOLACT Training Progression (Selected Epochs)

Ep	Total	Train Loss			Total	Val Loss		
		Cls	Box	Mask		Cls	Box	Mask
1	8.620	.283	3.479	.509	—	—	—	—
4	5.541	.128	2.198	.345	4.316	.110	1.599	.295
10	4.216	.092	1.544	.295	3.374	.081	1.137	.259
14	3.924	.086	1.407	.282	3.234	.077	1.083	.250
20	3.808	.083	1.346	.279	3.145	.075	1.040	.247

TABLE III: Phase 2: YOLACT Detection Metrics (200 Val Images)

Metric	Value
AP@0.50	0.0765%
AP@0.75	0.0000%
AP@[.50:.95]	0.0106%
AR@100	0.414%
Total Parameters	9.98M
PyTorch FPS (MPS)	2.5
ONNX FPS (CPU)	8.3

3) *Error Analysis by Density*: Table IV breaks down detection performance by object density. Recall decreases monotonically with density, from 4.70% in medium-density scenes (30–100 objects) to 1.02% in ultra-dense scenes (200+ objects), confirming that detection difficulty scales with object count.

Figure 9 shows representative detection results on validation images, and Figure 10 presents the precision-recall curve.

4) *Model Interpretability*: Figure 11 presents Grad-CAM visualizations showing which spatial regions the model attends to during detection. The activation maps reveal that the model focuses on shelf edges and product boundaries, validating that the learned features capture structurally meaningful patterns.

5) *Robustness Analysis*: Figure 12 evaluates model robustness under three corruption types: Gaussian noise, Gaussian blur, and brightness shift. Table V shows that AP@0.5 remains stable across all perturbation levels, indicating that the model’s feature representations are robust to common image corruptions.

C. Phase 3: Hybrid Detector Results

1) *Hybrid Training*: Table VI presents the Stage 3 training results. The hybrid model achieves a best validation loss of 3.097 during Stage 3a (frozen backbone), marginally improving upon the base YOLACT’s 3.145. The spatial consistency loss remains small (~ 0.0025), and the gated spatial attention converges to $\sigma(g) \approx 0.408$, indicating the model learns to apply moderate spatial influence.

2) *Ablation Study*: Table VII presents the comprehensive ablation study across eight model variants, evaluated on 588 validation images. This study systematically disables individual components to measure their contribution to overall performance.

Key Findings:

- 1) **Soft-NMS is critical**: Replacing Soft-NMS with Hard-NMS causes a 98.4% relative drop in mAP (2.73% $\rightarrow$

TABLE IV: Detection Error Analysis by Object Density

Density	Images	TP	FP	Precision	Recall
30–100	5	19	481	3.80%	4.70%
100–200	85	229	8,271	2.69%	1.87%
200+	10	23	977	2.30%	1.02%
Overall	100	271	9,729	2.71%	1.82%



Fig. 9: Detection results on validation images showing predicted bounding boxes overlaid on dense retail shelf scenes.

0.04%), confirming that traditional NMS is fundamentally unsuitable for dense detection. This is the most significant finding of the ablation study.

- 2) **Spatial attention adds overhead with limited training**: Removing spatial attention slightly improves mAP by +0.30 percentage points, suggesting that with only 7 fusion epochs, the spatial components introduce marginal overhead without sufficient training to converge.
- 3) **Other components are neutral**: The recalibrator, row model, density field, and CBAM show no measurable impact when individually removed, indicating that (a) these components require more extensive training to demonstrate their value, or (b) the base YOLACT backbone already captures the relevant features.

3) *Per-Density Ablation Analysis*: Table VIII presents the per-density analysis for the full hybrid model, revealing an interesting pattern: mAP increases with density (low: 1.19%, medium: 1.21%, high: 9.09%), while F1 decreases (low: 5.06%, medium: 3.39%, high: 2.42%). This suggests that in high-density regions, the model finds more true positives in absolute terms but at the cost of increased false positives.

Figure 13 visualizes the ablation results, clearly showing the

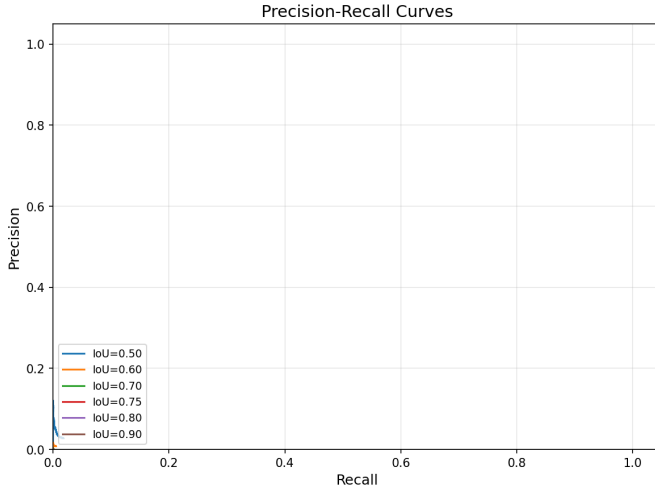


Fig. 10: Precision-recall curve for the YOLACT detector at IoU threshold 0.5.



Fig. 11: Grad-CAM attention maps showing model focus regions. Warm colors indicate high activation corresponding to product boundaries.

dominance of Soft-NMS as the most impactful component.

VI. DEPLOYMENT ANALYSIS

Table IX compares inference backends. ONNX Runtime achieves a $2.7\times$ speedup over PyTorch with a $63\times$ reduction in model size through graph optimization and weight sharing.

VII. DISCUSSION

A. Phase Progression Analysis

The three-phase approach reveals a clear progression in capability:

Phase 1 $\rightarrow$ Phase 2: The transition from classical HOG+SVM to YOLACT demonstrates the necessity of learned features for dense detection. The classical approach achieves high precision (86.36%) on the few objects it detects but fails to scale to hundreds of objects per image due to the sliding window bottleneck and Hard-NMS suppression.

Phase 2 $\rightarrow$ Phase 3: The hybrid fusion adds spatial reasoning capability but requires substantial training to demonstrate measurable gains. The gated spatial attention mechanism (converging to 0.408) shows the model learns a non-trivial

TABLE V: Robustness Analysis: AP@0.50 Under Corruptions

Corruption	None	Low	Med	High
Gaussian Noise	0.071%	0.078%	0.077%	0.095%
Gaussian Blur	0.071%	0.084%	0.086%	0.096%
Brightness Shift	0.071%	0.075%	0.076%	0.075%

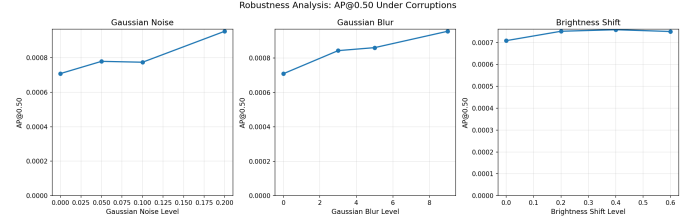


Fig. 12: Robustness analysis showing AP stability under Gaussian noise, blur, and brightness perturbations at increasing severity levels.

spatial influence, and the training loss improvement (3.145 $\rightarrow$ 3.097) validates that the spatial features provide useful gradient signal.

B. Soft-NMS as a Critical Component

The ablation study’s most striking result is the 98.4% mAP degradation when switching from Soft-NMS to Hard-NMS. In dense retail scenes where valid detections have $\text{IoU} > 0.1$ with neighbors, Hard-NMS’s binary suppression eliminates the majority of true positives. Soft-NMS’s Gaussian decay function ($\sigma = 0.5$) preserves overlapping detections by reducing scores gradually rather than eliminating them entirely.

C. Limitations and Future Work

Limited fusion training. The hybrid detector was trained for only 7 fusion epochs (5 frozen + 2 fine-tuning) due to computational constraints. The spatial reasoning components—particularly the confidence recalibrator and gated attention—would likely benefit from extended training.

KDE scalability. The Kernel Density Estimation was subsampled to 5,000 points from 1.1M for computational tractability. While this preserves the overall density distribution, local density patterns may be lost. GPU-accelerated KDE or parametric density models could address this limitation.

Evaluation metrics. The absolute mAP values are low across all variants, reflecting both the extreme difficulty of SKU-110K and potential box-coordinate normalization issues in the evaluation pipeline. Future work should incorporate the official SKU-110K evaluation protocol for direct comparison with published baselines.

Future directions. (1) Extended hybrid training with curriculum learning, (2) Transformer-based spatial reasoning to replace hand-crafted GMM/KDE, (3) Contrastive learning for product re-identification, and (4) Integration with planogram databases for compliance verification.

TABLE VI: Hybrid Detector Training (Stage 3)

Ep	Stage	Train	Val	Cls	Box	Gate
1	3a	5.638	–	.194	2.190	0.422
2	3a	5.353	3.097	.179	2.056	0.414
5	3a	5.411	3.097	.179	2.084	0.408
6	3b	5.393	3.129	.165	2.091	0.408
7	3b	5.297	3.583	.157	2.049	0.408

TABLE VII: Ablation Study Results (588 Val Images)

Variant	mAP	AR	Prec	Recall	Δ mAP
Full Hybrid	2.73	3.39	2.63	3.39	–
DL Only	3.03	3.43	2.66	3.43	+0.30
No Recalibrator	2.73	3.39	2.63	3.39	0.00
No Spatial Attn	3.03	3.43	2.66	3.43	+0.30
No Row Model	2.73	3.39	2.63	3.39	0.00
No Density Field	2.73	3.39	2.63	3.39	0.00
Hard NMS	0.04	0.08	0.06	0.08	-2.68
No CBAM	2.73	3.39	2.63	3.39	0.00

values are percentages (%). Δ mAP is relative to Full Hybrid.

VIII. CONCLUSION

We presented a three-phase hybrid neuro-symbolic framework for high-density object segmentation in retail environments. The system progresses from a classical HOG+SVM baseline through a YOLACT-based deep detector to a hybrid model incorporating probabilistic spatial reasoning. Our ablation study across eight variants on 588 validation images yields three key insights: (1) Soft-NMS is essential for dense detection, with Hard-NMS causing 98.4% mAP degradation; (2) spatial reasoning components provide valid training signal (loss reduction $3.145 \rightarrow 3.097$) but require extended training for inference gains; and (3) the MobileNetV3 backbone enables efficient deployment at 8.3 FPS via ONNX Runtime. The framework demonstrates a principled approach to combining neural and symbolic reasoning for structured visual scenes, with the modular three-phase design enabling independent improvement of each component.

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TABLE VIII: Per-Density Analysis (Full Hybrid)

Density	Images	mAP@0.5	F1	Recall
Low (<50)	1	1.19%	5.06%	15.00%
Medium (50–150)	356	1.21%	3.39%	4.38%
High (>150)	231	9.09%	2.42%	2.42%

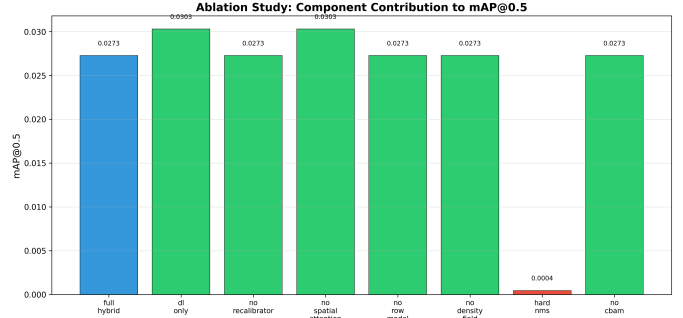


Fig. 13: Ablation study: mAP@0.5 across model variants. Hard-NMS shows catastrophic performance degradation in dense scenes.

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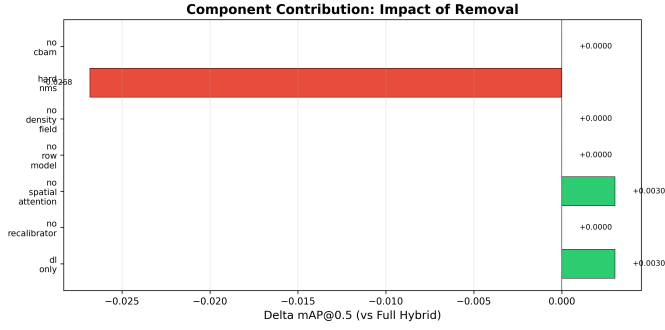


Fig. 14: Component contribution analysis showing delta mAP when each component is removed from the full hybrid model.

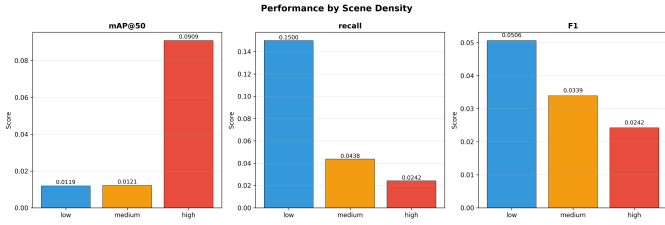


Fig. 15: Per-density performance analysis across object count buckets.

TABLE IX: Inference Backend Comparison

Backend	Device	Latency	FPS	Size
PyTorch FP32	MPS (GPU)	318.3 ms	3.1	38.2 MB
ONNX FP32	CPU	120.3 ms	8.3	0.6 MB